



Institute of Computing

CAPSTONE PROJECT OUTLINE - ACCEPTANCE FORM

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This letter confirms that the Objectives, Scope and Limitations, Functional and Non-functional requirements, and the hundred percent (100%) User Interface with thirty percent (30%) Functionality of the project entitled **SMARTALYZE: AN AI-ASSISTED DATA VISUALIZATION AND ANALYSIS PLATFORM** by **May Antonette Alterado, Donna Meg S. Eran & Gean T. Raz – Bachelor of science and Information Technology** has been accepted and approved by the respective technical panel committee members based on the previous Outline defense last **JUNE 02, 2026**.

The following specific objectives, scope and limitations, and functional and non-functional requirements are as follows:

Objectives:

1. Develop a web-based data ingestion module that enable users to upload structured datasets in CSV, Excel, and JSON formats, as well as image-based documents containing tabular data (e.g., PNG and JPG) for automated data extraction.
2. Implement an automated data-cleaning engine that utilizes rule-based techniques to detect and address common data-quality issues by:
 - 2.1. Imputing or removing missing values;
 - 2.2. Identifying and removing duplicate records; and
 - 2.3. Standardizing inconsistent data types.
3. Implement basic machine-learning techniques that provide intelligent recommendations for data structuring, grouping, summarization, and trend identification.
4. Design an interactive visualization dashboard that generates summarized insights and enables users to interpret data trends and patterns through graphical representations.
5. Develop a data export module that allows users to download cleaned and processed datasets in CSV, Excel, and JSON formats for academic, research, and business applications.

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Scope and Limitation

This study will focus on the development and evaluation of Smartalyze, an AI-assisted, web-based platform for data cleaning, visualization, and basic analysis. It is intended for users who have limited technical knowledge and need a simpler way to prepare, examine, and understand structured datasets. The platform will support multiple registered users, with each user having a separate account for managing uploaded files and saved work.

Smartalyze will accept structured data in CSV, Microsoft Excel (.xlsx and xls), and JSON formats. It will also allow users to upload PNG and JPG images containing tabular information through Optical Character Recognition. Before processing begins, the system will check whether the uploaded file is readable and arranged in a suitable table format.

The platform will include basic file dataset management functions. Users will be able to upload, rename, edit, organize, and delete datasets. They may also save unfinished work as a draft and continue editing it later. During data preparation, users will be allowed to add, rename, or remove columns as needed.

For data cleaning, the system will detect common issues such as missing values, duplicated records, and inconsistent data types using rule-based methods. It will also provide suggested values when relationships among fields can be identified. For instance, if a city is already indicated in the dataset, the system may suggest a corresponding ZIP code for a missing field. These suggestions will not be applied automatically and will still require user confirmation.

After the dataset has been prepared, Smartalyze will provide basic analytical functions such as descriptive statistics, grouping, summarization, linear regression for identifying trends, and interquartile range-based detection of possible anomalies. The system will also generate charts, tables, and AI-assisted explanations to help users understand the results. A progress indicator will be shown while data is being processed. Users will also be able to download cleaned datasets and selected analytical outputs in the formats supported by the platform.

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The study will be limited to structured datasets and images that contain readable tabular information. The system will not be designed to process handwritten content, audio, video, unstructured documents, or large collections of free-form text. The quality of OCR results will depend on factors such as image clarity, lighting, and the readability of the source material.

Suggestions for missing values will depend on the information already available in the dataset and the relationships that can be identified among its fields. Because of this, the suggested values may not always be correct and should be reviewed by the user. Likewise, the explanations and analytical outputs produced by the system will only serve as support for interpretation and will not replace professional statistical analysis or expert judgement.

The visualization features will also be limited to the chart and table types included in the platform. Smartalyze will not provide advanced business intelligence dashboards, real-time analytics, or predictive analysis beyond the methods implemented in the system. Uploaded files will also be subject to the file-size limit set by the platform to maintain reliable performance and efficient processing.

Functional and Non-Functional Requirements

The Smartalyze system shall provide the following functional requirements to support automated data preparation, visualization, and basic analysis for its intended users:

- 1. User Authentication and Account Management.** Users can register, log in, log out, and manage their own account. Each user can only access and manage their own datasets.
- 2. Dataset Upload and File Management.** Users can upload CSV, XLS, XLSX, JSON, PNG, or JPG files, which are validated and size-checked before processing. Users can organize and delete their uploaded datasets.
- 3. OCR Processing.** The system extracts tabular data from uploaded PNG/JPG images via OCR and converts it into an editable structured dataset.

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4. Dataset Editing. Users can edit cell values, add or remove columns, and rename column headers. Draft-style saving of unfinished work for later continuation is planned as an enhancement to the current versioning workflow.

5. Data Cleaning. The system detects missing values, duplicates, inconsistent types/casing/white-space, and invalid dates, presenting cleaning suggestions for user review before applying them.

6. Relationship-Based Data Recommendations. The system suggests missing values by detecting consistent patterns between related columns, requiring user confirmation before any suggestion is applied.

7. Data Analysis. The system provides descriptive statistics, grouping/summarization, linear-regression trend analysis, IQR-based anomaly detection, and AI-assisted explanations of the results.

8. Data Visualization. The system renders bar, line, pie, and area charts, plus histograms and frequency distributions, limited to compatible column types.

9. Progress Monitoring. The system shows a progress indicator during uploads, OCR, cleaning, analysis, visualization, and export.

10. Data Export. Users can download cleaned datasets and selected analytical outputs in the system's supported formats.

11. Dataset Version Management. The system keeps prior dataset versions and lets users restore an earlier version at any time.

The Smartalyze system shall satisfy the following non-functional requirements to ensure its quality, reliability, security, and usability and overall performance.

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1. Performance Efficiency. The system shall process datasets efficiently, showing progress indicators during long operations (upload, OCR, cleaning, analysis, export). Files are capped at 20 MB, with oversized uploads rejected and flagged to the user.

2. Reliability. The system shall preserve data integrity across all dataset operations and remain accessible to authorized users whenever the server is running (Availability), with saved datasets and versions persisting until the user modifies or deletes them.

3. Security. The system shall require authentication for all dataset and analytics access, store passwords encrypted, and isolate each user's data so only its owner can view or modify it.

4. Interaction Capability. The system shall provide a simple, guided interface usable by people with limited technical background, with clear instructions, validation messages, and error feedback throughout.

5. Maintainability. The system shall use a modular architecture so file formats, cleaning methods, visualizations, and analytical features can be extended without major rework elsewhere in the system.

6. Flexibility. The system shall support multiple concurrent users and a growing number of datasets while maintaining acceptable performance (Scalability), and shall be adaptable to future formats or deployment needs.

7. Compatibility. The system shall run in modern browsers (Chrome, Edge, Firefox, Safari) without requiring any additional software installation.

8. Functional Suitability. The system's generated outputs OCR extractions, statistics, trends, anomalies, recommendations, and AI explanations shall correctly reflect the uploaded data's quality and completeness, and shall require user review before being acted on.

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Noted by:


ENGR. JONILYN T. DABALOS, MIT

Adviser

Date Signed: 7/30/2026

Conformed by:

ANALYN V. CASIPE, MIT

Panel Chairman

Date Signed: _____

JOVITO P. BOLACROY JR., MIS

Panel Member

Date Signed: _____

IAN JAY PADIOS

Panel Member

Date Signed: _____

Accepted as Capstone Project Title for the program BS in **[Information Technology/Systems]**.

Signed by:

REBAN CLIFF A. FAJARDO, MIT

[BSIT/BSIS Program Chairperson]

Date Signed: _____

MARK VAN M. BULADACO, DBMIS

Dean, Institute of Computing

Date Signed: _____

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